جامعة الإمام عبد الرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

ARTI - 309
Mathematical Foundations for Artificial Intelligence

2025-2026 - Term 2

Lectures 5&6

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Outline

Partial Derivatives of Multivariate Functions
James Stewart – Chapter 14, Section 3

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14.3 Partial Derivatives

Concept Explanation.

The heat index (humidex) I is the perceived air temperature when the actual temperature is T and the relative humidity is H.

$I = f(T, H)$.

Relative humidity (%)

Actual temperature (°F)

T \ H	50	55	60	65	70	75	80	85	90
90	96	98	100	103	106	109	112	115	119
92	100	103	105	108	112	115	119	123	128
94	104	107	111	114	118	122	127	132	137
96	109	113	116	121	125	130	135	141	146
98	114	118	123	127	133	138	144	150	157
100	119	124	129	135	141	147	154	161	168

14.3 Partial Derivatives

If you look at the values in one row only, it shows a function of one variable only, the humidity.
If you look at the values in one column only, it shows a function of also one value only, the temperature.

You can think of the derivative along that row or that column.

Relative humidity (%)

Actual temperature (°F) T \ H	50	55	60	65	70	75	80	85	90
90	96	98	100	103	106	109	112	115	119
92	100	103	105	108	112	115	119	123	128
94	104	107	111	114	118	122	127	132	137
96	109	113	116	121	125	130	135	141	146
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14.3 Partial Derivatives

Relative humidity (%)

Actual temperature (°F) T \ H	50	55	60	65	70	75	80	85	90
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Look at the blue column.

You can define the function: $g(T) = f(T, 70)$.

$g(T)$ shows how the humidex increases as the actual temperature increases when the relative humidity is constant at 70%.

Now evaluate the change of humidex around the temperature $T = 96°\text{F}$

The derivative of g when $T = 96°\text{F}$ is the rate of change of I with respect to T when $T = 96°\text{F}$:

$$g'(96) = \lim_{h \to 0} \frac{g(96 + h) - g(96)}{h} = \lim_{h \to 0} \frac{f(96 + h, 70) - f(96,70)}{h}$$

Now approximate $g'(96)$ by taking $h = \pm 2$

14.3 Partial Derivatives

$$g'(96) \approx \frac{g(98)-g(96)}{2} = \frac{f(98,70)-f(96,70)}{2} = \frac{133-125}{2} = 4.$$

$$g'(96) \approx \frac{g(94)-g(96)}{-2} = \frac{f(94,70)-f(96,70)}{-2} = \frac{118-125}{-2} = 3.5.$$

Relative humidity (%)

Actual temperature (°F)

T \ H	50	55	60	65	70	75	80	85	90
90	96	98	100	103	106	109	112	115	119
92	100	103	105	108	112	115	119	123	128
94	104	107	111	114	118	122	127	132	137
96	109	113	116	121	125	130	135	141	146
98	114	118	123	127	133	138	144	150	157
100	119	124	129	135	141	147	154	161	168

Now take the average:

$$g'(96) = \frac{4+3.5}{2} = 3.75.$$

This means that when the actual temperature is 96 °F and the relative humidity is 70%, the apparent temperature rises by about 3.75 °F for every degree °F that the actual temperature rises.

14.3 Partial Derivatives

Actual temperature (°F) \ Relative humidity (%) H	50	55	60	65	70	75	80	85	90
90	96	98	100	103	106	109	112	115	119
92	100	103	105	108	112	115	119	123	128
94	104	107	111	114	118	122	127	132	137
96	109	113	116	121	125	130	135	141	146
98	114	118	123	127	133	138	144	150	157
100	119	124	129	135	141	147	154	161	168

Next, approximate the row derivative for T=96 °F at H=70%.

G(H) $=$ $f(96, H)$ describes how the humidex increases as the relative humidity H increases, for a constant temperature T.

The derivative is G'(70) $= \lim_{h\to 0} \frac{G(70+h)-G(70)}{h} = \lim_{h\to 0} \frac{f(96,70+h)-f(96,70)}{h}$

Now approximate the value of the row derivative with $h = \pm 5$.

On the right side: G'(70) $\approx \frac{G(75)-G(70)}{5} = \frac{f(96,75)-f(96,70)}{5} = \frac{130-125}{5} = 1.$

On the left side: G'(70) $\approx \frac{G(65)-G(70)}{-5} = \frac{f(96,65)-f(96,70)}{-5} = \frac{121-125}{-5} = 0.8.$

And the average is : $G'(70) = \frac{1+0.8}{2} = 0.9.$

Which means that for temperature T=96 °F and H=70%, the humidex rises approximately with 0.9 °F for every percent that the relative humidity rises.

Relative humidity (%)

Actual temperature (°F) T \ H	50	55	60	65	70	75	80	85	90
90	96	98	100	103	106	109	112	115	119
92	100	103	105	108	112	115	119	123	128
94	104	107	111	114	118	122	127	132	137
96	109	113	116	121	125	130	135	141	146
98	114	118	123	127	133	138	144	150	157
100	119	124	129	135	141	147	154	161	168

14.3 Partial Derivatives

Conclusion: The humidex rises more with one degree of the temperature than with one percent of the relative humidity.

14.3 Partial Derivatives

Definition: If f is a function of two variables, the partial derivative of f with respect to x at (a, b) is denoted by $f_x(a, b)$ and the partial derivative of f with respect to y at (a, b) is denoted by $f_y(a, b)$. They are defined by:

$$f_x(a, b) = \lim_{h \to 0} \frac{f(a+h,b) - f(a,b)}{h}$$

$$f_y(a, b) = \lim_{h \to 0} \frac{f(a,b+h) - f(a,b)}{h}$$

Notations: Let $z = f(x, y)$, then there are several notations for the partial derivative:

$f_x(x, y) = \frac{\partial}{\partial x} f(x, y)$, and without arguments: $f_x = \frac{\partial f}{\partial x} = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f$

$f_y(x, y) = \frac{\partial}{\partial y} f(x, y)$, and without arguments: $f_y = \frac{\partial f}{\partial y} = \frac{\partial z}{\partial y} = f_2 = D_2 f = D_y f$

14.3 Partial Derivatives

Example 1: Compute the partial derivatives of the function

$$f(x, y) = x^3 + x^2y^3 - 2y^2.$$

Compute the value for $f_x(2, 1)$ and $f_y(2, 1)$.

Solution:

For f_x consider y to be constant: $f_x(x, y) = 3x^2 + 2xy^3 - 0 = 3x^2 + 2xy^3$

$f_x(2, 1) = 3 * 4 + 2 * 2 * 1 = 12 + 4 = 16.$

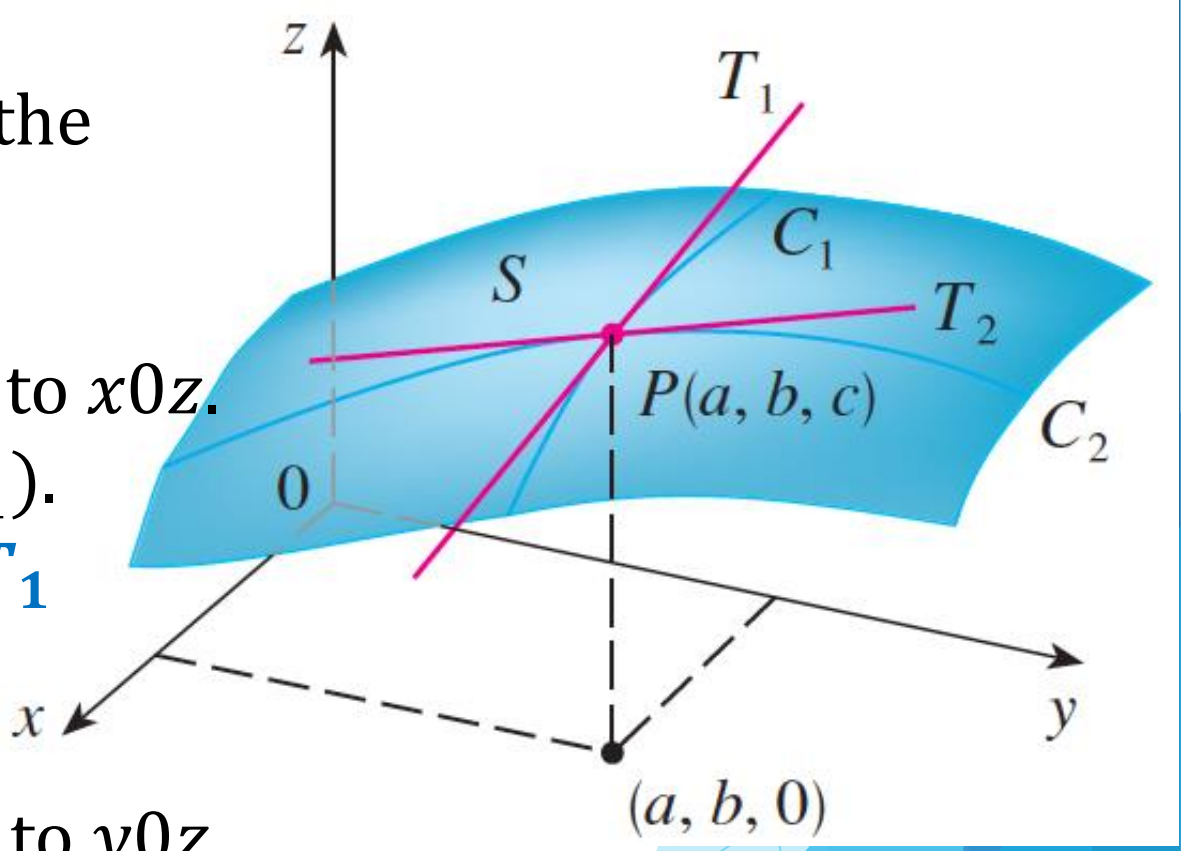
For f_y consider x to be constant:

$f_y(x, y) = 0 + x^2 * 3y^2 - 2 * 2y = 3x^2y^2 - 4y$

$f_y(2, 1) = 3 * 4 * 1 - 4 * 1 = 8.$

14.3 Partial Derivatives – Geometric Interpretation

The equation $z = f(x, y)$ is a surface S.
If $f(a, b) = c$, then $P(a, b, c)$ is a point on the surface S.
By fixing $y = b$, we get the curve
$C_1: z = f(x, b)$ which is in a plane parallel to $x0z$.
(The vertical plane $y = b$ intersects S in C_1).
$f_x(a, b)$ is the slope of the tangent line T_1 to the curve C_1 at the point P.
By fixing $x = a$, we get the curve
$C_2: z = f(a, y)$ which is in a plane parallel to $y0z$.
(The vertical plane $x = a$ intersects S in C_2).
$f_y(a, b)$ is the slope of the tangent line T_2 to the curve C_2 at the point P.



14.3 Partial Derivatives

Example 2: If $f(x, y) = 4 - x^2 - 2y^2$, find $f_x(1, 1)$ and $f_y(1, 1)$ and interpret these numbers as slopes.

Solution:

$f_x(x, y) = -2x$ ➔ $f_x(1, 1) = -2$

$f_y(x, y) = -4y$ ➔ $f_y(1, 1) = -4$

$z = 4 - x^2 - 2y^2$ is a paraboloid.

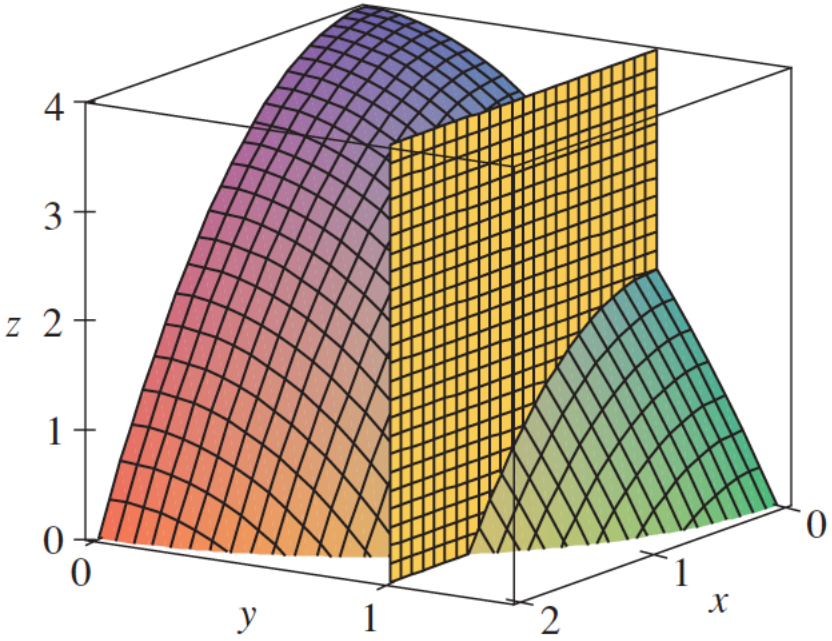
Note that $f(1, 1) = 4 - 1 - 2 = 1$ ➔ P(1,1,1) is the point on the surface where we calculated the partial derivatives.

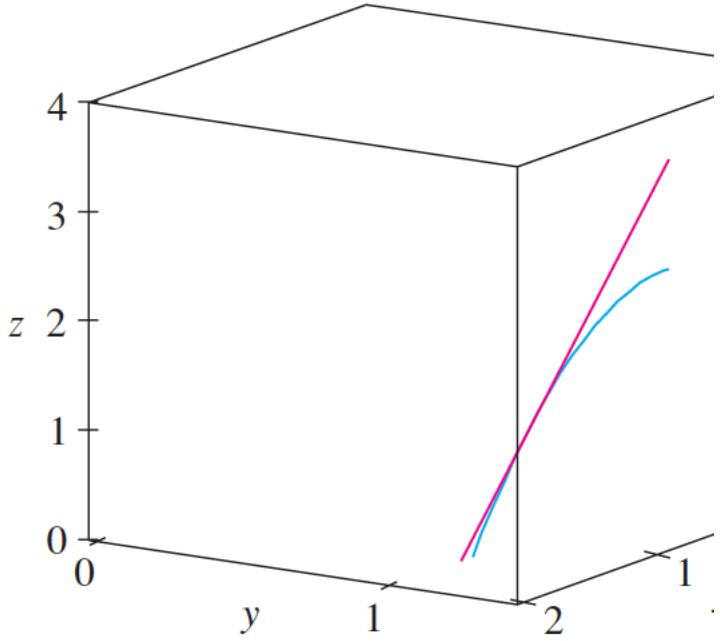
14.3 Partial Derivatives

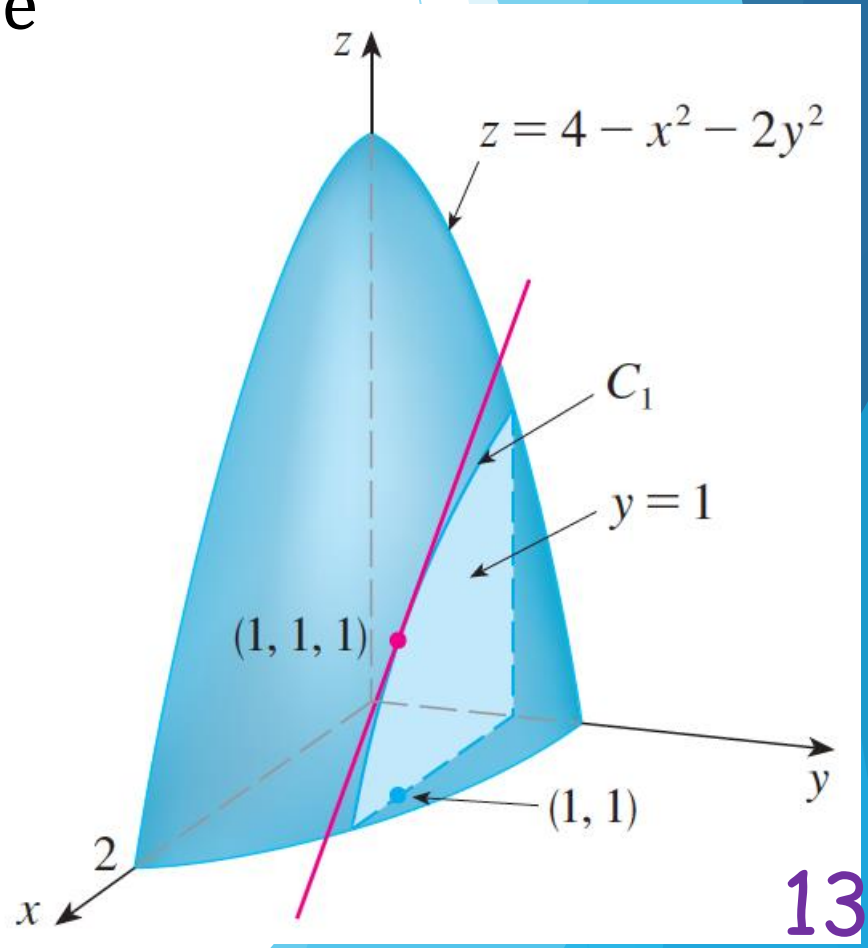
Example 2 - continued:

The plane $y = 1$ intersects $z = 4 - x^2 - 2y^2$ in the parabola $z = 2 - x^2$.

C_1 is this parabola. The pink line is tangent with the Slope equal to the derivative $f_x(1, 1) = -2$.





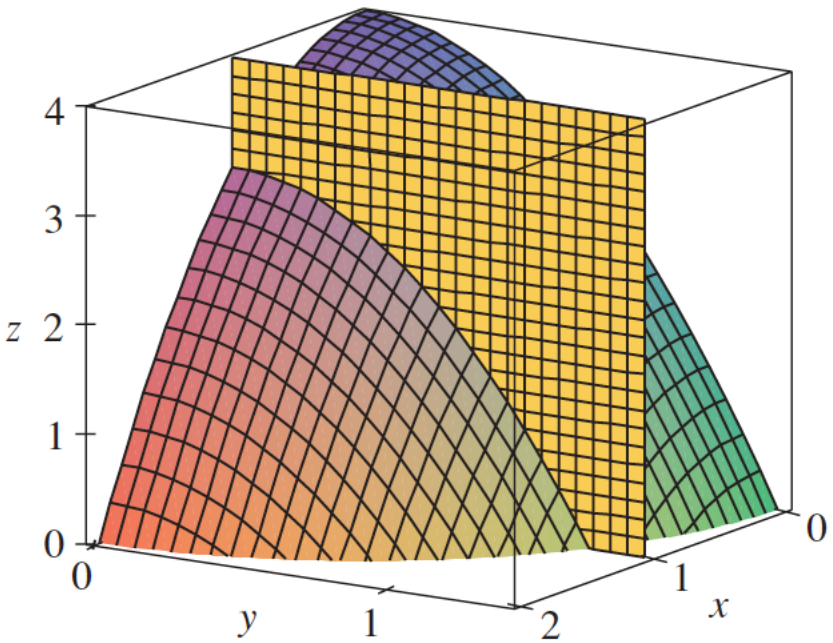


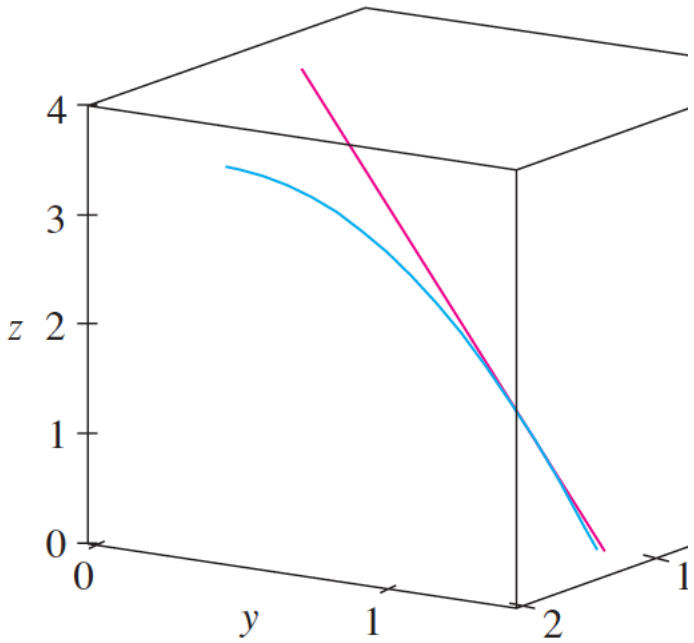
14.3 Partial Derivatives

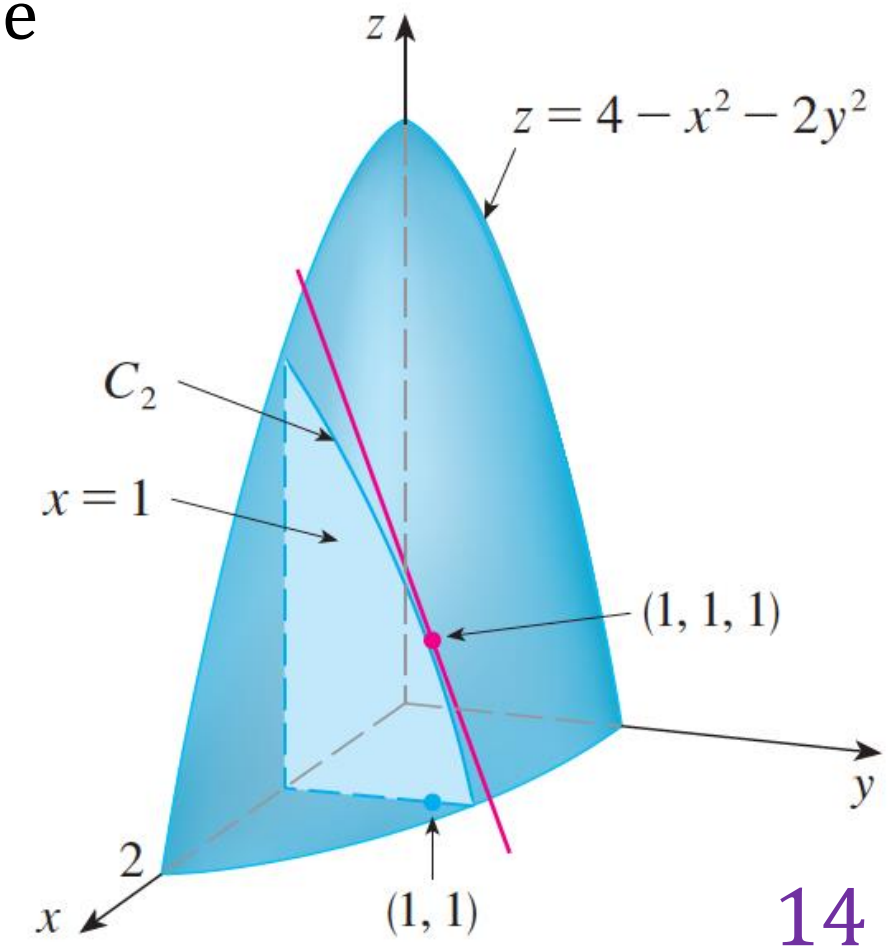
Example 2 - continued:

The plane $x = 1$ intersects $z = 4 - x^2 - 2y^2$ in the parabola $z = 3 - 2y^2$.

C_2 is this parabola. The pink line is tangent with the Slope equal to the derivative $f_y(1,1) = -4$.







Work on practice problems: 11 and 12 page 936

14.3 Partial Derivatives

Example 3: If $f(x,y) = \sin\left(\frac{x}{1+y}\right)$, calculate $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

Solution: Using the Chain Rule for functions of one variable, we have

$$\frac{\partial f}{\partial x}(x,y) = \cos\left(\frac{x}{1+y}\right)\frac{\partial}{\partial x}\left(\frac{x}{1+y}\right) = \cos\left(\frac{x}{1+y}\right)\frac{1}{1+y}$$

$$\frac{\partial f}{\partial y}(x,y) = \cos\left(\frac{x}{1+y}\right)\frac{\partial}{\partial y}\left(\frac{x}{1+y}\right) = \cos\left(\frac{x}{1+y}\right)\left(-\frac{x}{(1+y)^2}\right)$$

$$= -\cos\left(\frac{x}{1+y}\right)\frac{x}{(1+y)^2}$$

14.3 Partial Derivatives

Example 4: Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$, if z is defined implicitly as a function of x and y by the equation

$$x^3 + y^3 + z^3 + 6xyz = 1$$

Solution: For $\frac{\partial z}{\partial x}$, differentiate with respect to x, and keep y constant.

$$3x^2 + 6yz + 3z^2\frac{\partial z}{\partial x} + 6xy\frac{\partial z}{\partial x} = 0$$

Now solve the equation:

$$\frac{\partial z}{\partial x} = -\frac{x^2 + 2yz}{z^2 + 2xy}$$

And for $\frac{\partial z}{\partial y}$, we have $\frac{\partial z}{\partial y} = -\frac{y^2+2xz}{z^2+2xy}$

14.3 Partial Derivatives – More Variables

Example 5: Find f_x , f_y , and f_z if $f(x, y, z) = e^{xy} \ln z$.

Solution:

For f_x, keep y and z constant: $f_x(x, y, z) = ye^{xy} \ln z$

For f_y, keep x and z constant: $f_y(x, y, z) = xe^{xy} \ln z$

For f_z, keep x and y constant: $f_z(x, y, z) = \dfrac{e^{xy}}{z}$

Here we need a four dimensional space to plot the graph… Hmm.

https://www.matrixlab-examples.com/simple-animation-3d.html

14.3 Partial Derivatives – Higher Derivatives

Definition: Consider the function of two variables $z = f(x, y)$. The following higher derivatives can be defined.

$$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

$$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 z}{\partial y \partial x}$$

$$(f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 z}{\partial x \partial y}$$

$$(f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

14.3 Partial Derivatives – Higher Derivatives

Example 6: Find the second partial derivatives of $f(x,y) = x^3 + x^2y^3 - 2y^2$.

Solution:

What is the degree of this polynomial? FIVE.

$f_x(x,y) = 3x^2 + 2xy^3$ and $f_y(x,y) = 3x^2y^2 - 4y$

Now $f_{xx} = \frac{\partial}{\partial x}(3x^2 + 2xy^3) = 6x + 2y^3$

$f_{xy} = \frac{\partial}{\partial y}(3x^2 + 2xy^3) = 6xy^2$

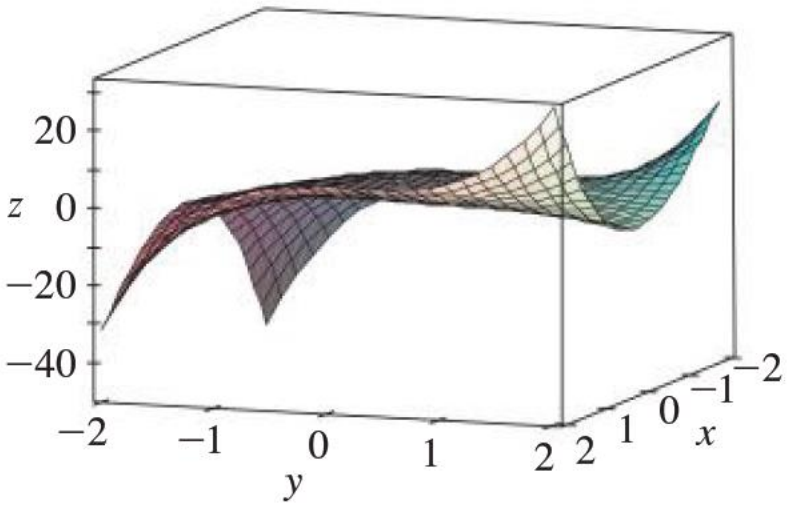
$f_{yx} = \frac{\partial}{\partial x}(3x^2y^2 - 4y) = 6xy^2$ *Notice that $f_{xy} = f_{yx}$*

$f_{yy} = \frac{\partial}{\partial y}(3x^2y^2 - 4y) = 6x^2y - 4$

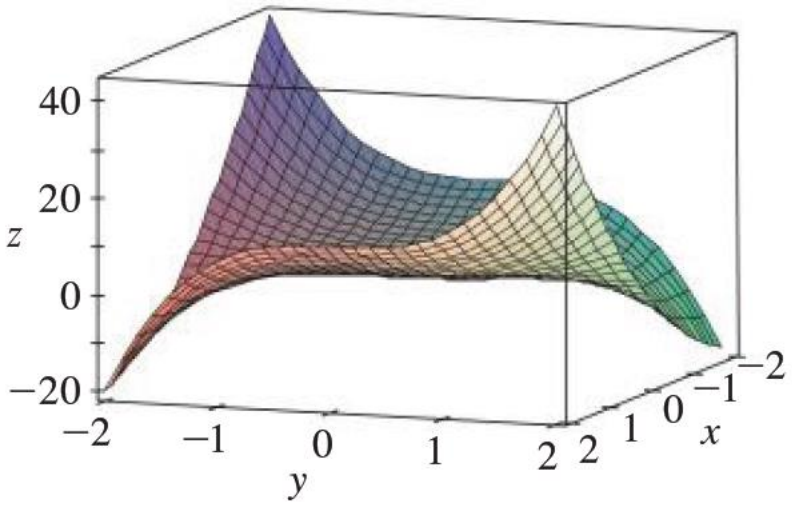
14.3 Partial Derivatives – Higher Derivatives

Example 6 - continued:

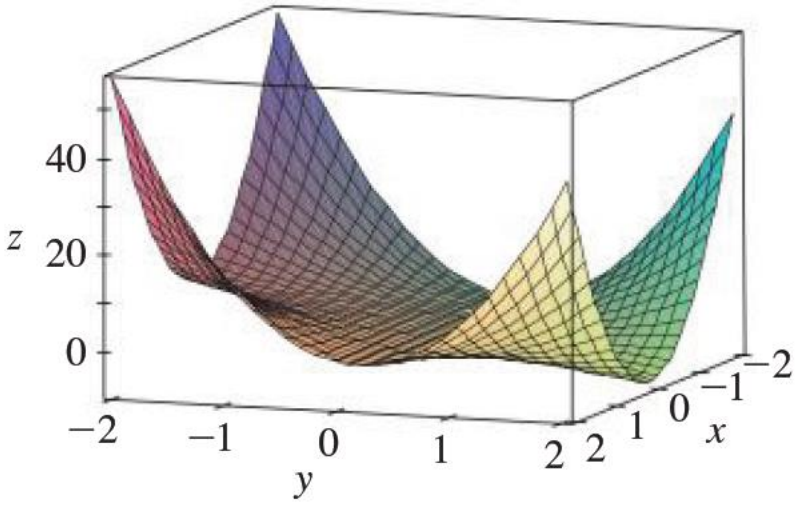
Now look at the graphs in CalcPlot3D.



f



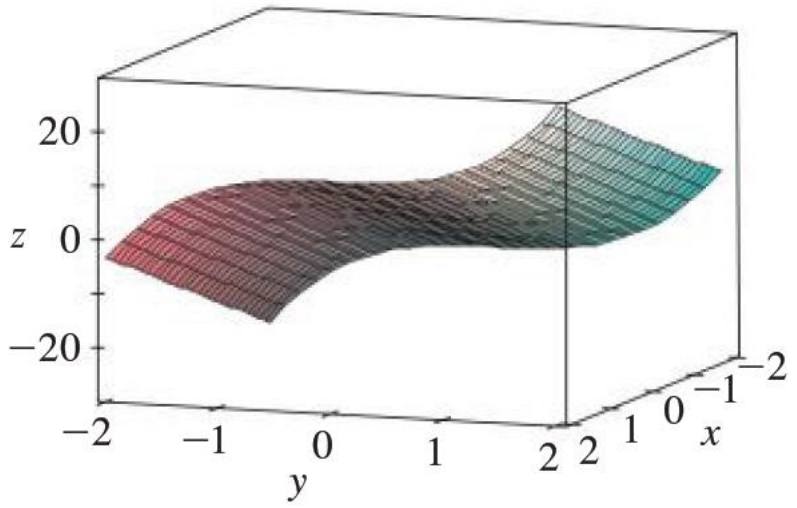
f_x



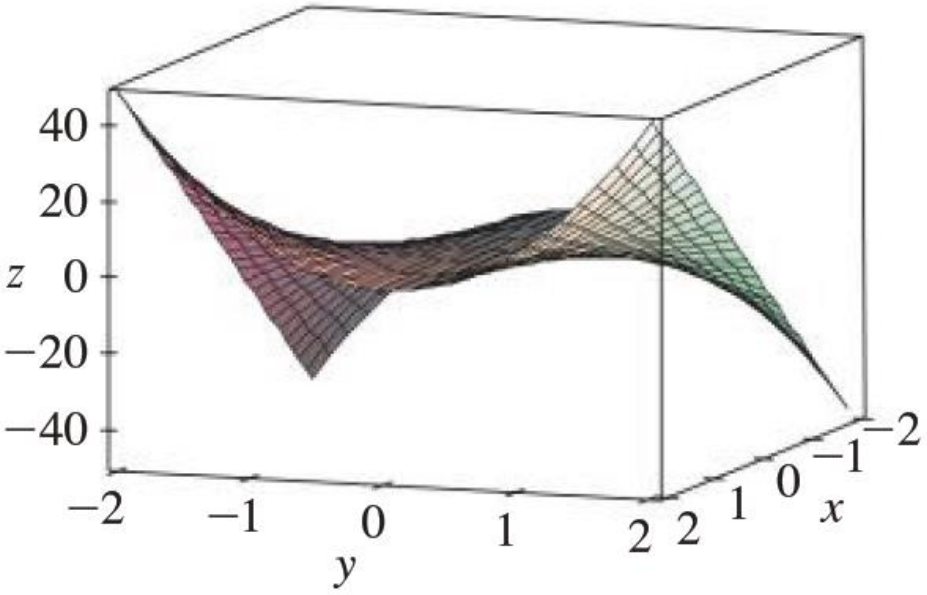
f_y

14.3 Partial Derivatives – Higher Derivatives

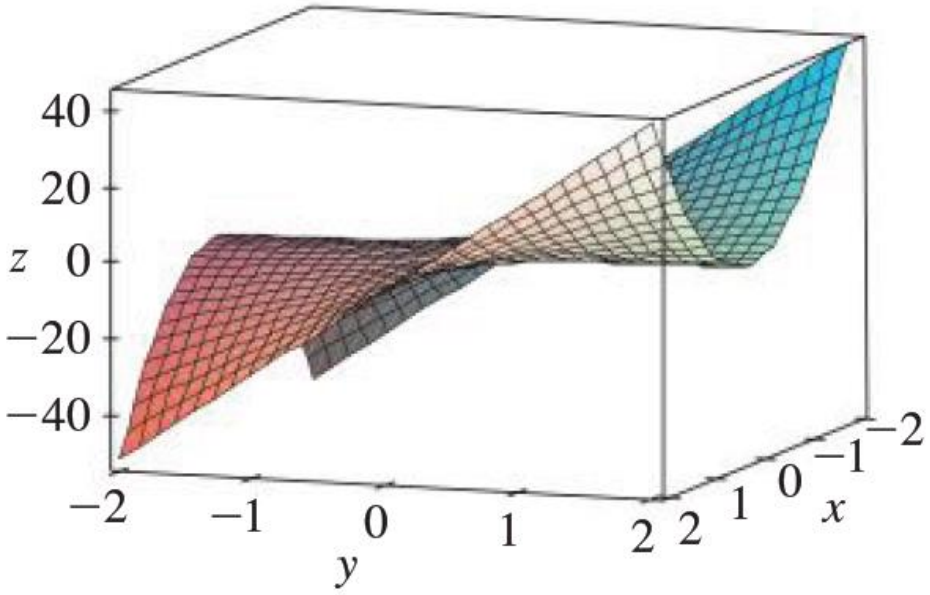
Example 6 - continued:



f_{xx}



$f_{xy} = f_{yx}$



f_{yy}

14.3 Partial Derivatives – Clairaut's Theorem

Clairaut's Theorem

Suppose f is defined on a disk D that contains the point (a, b). If the functions $\boldsymbol{f_{xy}}$ **and** $\boldsymbol{f_{yx}}$ **are both continuous** on D, then

$$\boldsymbol{f_{xy}(a, b) = f_{yx}(a, b)}$$

14.3 Partial Derivatives – Higher Order

Example 7: Calculate f_{xxyz} if $f(x, y, z) = \sin(3x + yz)$.

Solution: $f_{xxyz} = \left(\left((f_x)_x\right)_y\right)_z$

Chain rule: $f_x = 3\cos(3x + yz)$

Chain rule: $f_{xx} = -9\sin(3x + yz)$

Chain rule: $f_{xxy} = -9z\cos(3x + yz)$

Product rule: $f_{xxyz} = -9\cos(3x + yz) + 9yz\sin(3x + yz)$

Work on practice problems: 47-48-64-65-76a

We Finished Partial Derivatives 14.3 -Stewart

Next comes The Chain Rule